ClimateScope

-Visualising Global Weather Trends
and Extreme Events

Internship Domain: Data Visualisation
Academic Session: 2025

PROJECT OVERVIEW:
This report serves as the final documentation for the ClimateScope initiative. It details the end-to-end development of an interactive intelligence platform designed to decode the complexities of the Global Weather Repository. The project demonstrates a robust data engineering and analytics pipeline, starting with high-performance preprocessing and statistical validation within Google Colab. These refined datasets were then transitioned into Tableau to build a high-fidelity, interactive dashboard. This "Climate Scope" allows researchers and policy makers to filter global metrics, identify regional anomalies, and visualise environmental trends through a synchronised, data-driven interface.

Core Project Entities:

- **Project Supervisor:** Shanmukha Priya
- **Host Organisation:** Infosys Springboard
- **Intern Name:** Gollu Bhagya Sri Satya

ABSTRACT

The ClimateScope initiative represents a sophisticated analytical response to the increasing volatility of global atmospheric patterns, specifically designed to transform the high-velocity data within the Global Weather Repository into actionable environmental intelligence. This project documents a comprehensive, four-milestone data engineering and visualisation pipeline that bridges the gap between raw meteorological records and strategic decision-making. The technical journey commenced in the Google Colab environment, where the global observation dataset underwent a rigorous structural audit, multi-stage cleaning, and feature engineering to isolate critical variables, including particulate matter, UV radiation indices, and barometric pressure. Through the application of advanced statistical preprocessing—including outlier detection via Interquartile Range (IQR) logic and temporal synchronisation—the raw CSV data was refined into a high-integrity analytical base. This processed data was subsequently integrated into Tableau Desktop, where a "Bento Box" dashboard architecture was developed to facilitate multivariate exploration. The final interactive platform allows users to synchronise eight distinct visualisation modules, uncovering deep-seated correlations between solar intensity, industrial pollutants, and storm signatures. By providing a scalable, data-driven "Climate Scope," this project serves as a robust framework for researchers and environmental policy-makers to monitor regional anomalies, evaluate public health risks, and visualise the evolving state of the global climate in a synchronised, intuitive interface.

CHAPTER 1: INTRODUCTION

1.1 Project Background and Significance

The global climate landscape is currently undergoing a period of unprecedented volatility, characterised by localised extreme weather events that challenge traditional forecasting models. The ClimateScope project was initiated to bridge the gap between complex meteorological data repositories and human-centric visual intelligence. By utilising the Global Weather Repository, this project moves beyond simple temperature tracking to provide a multidimensional view of atmospheric health. The significance of this work lies in its ability to synthesise diverse metrics—such as Particulate Matter, Ultraviolet (UV) radiation, and wind gust velocity—into a single, synchronised interface. This allows for the identification of "Hazardous Corridors" where industrial pollution and high-intensity solar radiation intersect, providing a critical tool for environmental researchers and public health officials to monitor anomalies and safeguard urban populations.

1.2 Problem Statement

Current environmental reporting often suffers from "Data Siloing," where air quality, thermal trends, and wind patterns are analysed in isolation. This fragmented approach prevents a holistic understanding of how these variables interact—for instance, how high solar intensity acts as a catalyst for ground-level Ozone formation. Furthermore, raw datasets like the Global Weather Repository are too dense for immediate interpretation by non-technical stakeholders.

There is a clear necessity for a robust analytical pipeline using Google Colab to ingest and clean these millions of data points, transforming them into an interactive Tableau "Bento Box" dashboard. This allows for the transition from raw, fragmented records to a synchronised system that facilitates real-time exploration and anomaly detection across all weather and health metrics.

1.3 Scope of the Study

The scope of this project is defined by the four core milestones of the internship program. It covers the end-to-end lifecycle of a data project, beginning with the structural audit of the dataset in Google Colab and concluding with the deployment of a high-fidelity interactive prototype in Tableau. The study focuses on global urban centres, using daily-updated records spanning 2023 through 2025 to analyse 43 distinct variables. The boundaries of this report are strictly technical and analytical, focusing on the data-engineering logic required to purify the records and the visualisation principles used to communicate environmental risks effectively. This scope ensures a comprehensive transition from raw, unprocessed CSV data to a synchronised analytical platform.

1.4 Tools and Methodology

The methodology for ClimateScope is built on a "Dual-Engine" framework, utilising high-performance cloud computing for data engineering and an industry-leading business intelligence tool for interactive synthesis:

- **Google Colab (The Analytical Engine):** This environment was used for the "heavy lifting" of data engineering. It allowed for the application of Python libraries such as **Pandas** for complex data manipulation, **NumPy** for mathematical operations, and **Seaborn** for exploratory statistical sketching. This phase was critical for ensuring the dataset was "Tableau-ready" by programmatically handling null values and performing feature engineering to create new temporal dimensions, specifically extracting **Year** and **Month** for time-series analysis.
- **Tableau (The Visual Engine):** Used to build the final user-facing intelligence platform. Tableau's robust filtering engine and "Dashboard Actions" were leveraged to create a synchronized environment where multiple modules react simultaneously to user input. This facilitates a seamless "drill-down" experience, allowing stakeholders to transition from global weather overviews to granular, city-specific profiles across all 43 variables.

CHAPTER 2: MILESTONE 1 - DATA COLLECTION & PREPROCESSING

2.1 Objective of Milestone 1

The primary goal of this initial phase was to establish a high-quality data foundation for the ClimateScope platform.
The objectives included:

- Data Collection: Harvesting the global weather dataset from the designated repository.
- Structural Auditing: Analysing the dataset's architecture to identify key attributes and data types.
- Cleaning & Preprocessing: Implementing Python-based logic to remediate inconsistencies.
- Analytical Preparation: Ensuring the final output was optimised for downstream Key Performance Indicator (KPI) calculation and Tableau visualisation.

2.2 Dataset Overview

The project utilises the Global Weather Repository, a comprehensive dataset provided in CSV format. This repository contains global meteorological records characterised by several critical attributes:

- Geographic Identifiers: Location name and country.
- Atmospheric Metrics: Temperature, humidity, wind speed, barometric pressure, and cloud cover.
- Weather State: Qualitative condition text describing the environment.

2.3 Technical Execution: Data Ingestion

The data ingestion process was executed within a Google Colab environment using the Python Pandas library.

- **Functionality**: The read_csv() function was employed to load the raw file into a structured DataFrame.
- **Configuration**: The low_memory=False parameter was utilised during loading to ensure data integrity across large volumes of records.

	country	location_name	latitude	longitude	timezone	last_updated_epoch	last_updated	temperature_celsius	temperature_fahrenheit	condition_text	...	air_quality_PM2.5	air_quality_PM10	air_quality_us-epa-index	air_quality_gb-defra-index	sunrise	sunset	moonrise	moonset	moon_phase	moon_illumination
0	Afghanistan	Kabul	34.52	69.18	Asia/Kabul	1.715849e+09	16-05-2024 13:15	26.6	79.8	Partly Cloudy	...	8.4	26.6	1.0	1.0	4:50 AM	6:50 PM	12:12 PM	1:11 AM	Waxing Gibbous	55.0
1	Albania	Tirana	41.33	19.82	Europe/Tirane	1.715849e+09	16-05-2024 10:45	19.0	66.2	Partly cloudy	...	1.1	2.0	1.0	1.0	5:21 AM	7:54 PM	12:58 PM	2:14 AM	Waxing Gibbous	55.0
2	Algeria	Algiers	36.76	3.05	Africa/Algiers	1.715849e+09	16-05-2024 09:45	23.0	73.4	Sunny	...	10.4	18.4	1.0	1.0	5:40 AM	7:50 PM	1:15 PM	2:14 AM	Waxing Gibbous	55.0
3	Andorra	Andorra La Vella	42.50	1.52	Europe/Andorra	1.715849e+09	16-05-2024 10:45	6.3	43.3	Light drizzle	...	0.7	0.9	1.0	1.0	6:31 AM	9:11 PM	2:12 PM	3:31 AM	Waxing Gibbous	55.0
4	Angola	Luanda	-8.84	13.23	Africa/Luanda	1.715849e+09	16-05-2024 09:45	26.0	78.8	Partly cloudy	...	183.4	262.3	5.0	10.0	6:12 AM	5:55 PM	1:17 PM	12:38 AM	Waxing Gibbous	55.0

5 rows × 41 columns

2.4 Data Cleaning & Standardisation

To ensure the accuracy of the "Climate Scope," the raw data underwent a rigorous cleaning cycle:

- **Integrity Checks**: The dataset was audited for missing values and duplicates; notably, no duplicate records were identified in this phase.
- **Temporal Normalisation**: The last_updated column was converted from a string format to a formal datetime object using pd.to_datetime.
- **Text Standardisation**: Categorical data was normalised to ensure consistency in Tableau filters. Weather conditions were converted to lowercase, while country names were formatted into title case.

2.5 Feature Engineering

To facilitate time-series analysis in the final dashboard, new dimensional attributes were engineered from a standardised datetime column:

- **Temporal Extraction**: Specific columns for **Year**, **Month**, and **Day** were generated using the .dt accessor in Pandas. This allows the final user to filter the dashboard by specific periods to observe seasonal trends.

2.6 Output and Validation

The milestone concluded with a statistical validation of the numerical data using the describe() and info() methods to confirm the final structure. The result was a refined, cleaned dataset saved as a new CSV file, fully prepared for Exploratory Data Analysis (EDA) and visualisation.

	latitude	longitude	last_updated_epoch	last_updated	temperature_celsius	temperature_fahrenheit	wind_mph	wind_kph	wind_degree	pressure_r
count	123356.000000	123356.000000	1.233560e+05	123356	123356.000000	123356.000000	123356.000000	123356.000000	123356.000000	123356.0000(
mean	19.191767	21.994709	1.743297e+09	2025-03-30 03:25:01.515127296	21.665486	70.999629	8.072977	12.995777	169.263870	1014.0854:
min	-41.300000	-175.200000	1.715849e+09	2024-05-16 01:45:00	-29.800000	-21.600000	2.200000	3.600000	1.000000	947.0000(
25%	3.750000	-6.836100	1.729588e+09	2024-10-22 14:37:30	16.300000	61.300000	4.000000	6.500000	80.000000	1010.0000(
50%	17.250000	23.278350	1.743327e+09	2025-03-30 11:30:00	24.100000	75.400000	6.900000	11.200000	162.000000	1014.0000(
75%	40.400000	50.580000	1.756973e+09	2025-09-04 12:00:00	28.100000	82.600000	11.000000	17.600000	256.000000	1018.0000(
max	64.150000	179.220000	1.770710e+09	2026-02-10 21:00:00	49.200000	120.600000	1841.200000	2963.200000	360.000000	3006.0000(
std	24.427080	65.792350	1.583412e+07	NaN	9.547590	17.185514	7.391056	11.891706	103.421709	10.6279:

8 rows × 34 columns

```
0   country                       123356 non-null  object
1   location_name                 123356 non-null  object
2   latitude                      123356 non-null  float64
3   longitude                     123356 non-null  float64
4   timezone                      123356 non-null  object
5   last_updated_epoch            123356 non-null  int64
6   last_updated                  123356 non-null  datetime64[ns]
7   temperature_celsius           123356 non-null  float64
8   temperature_fahrenheit        123356 non-null  float64
9   condition_text                123356 non-null  object
10  wind_mph                      123356 non-null  float64
11  wind_kph                      123356 non-null  float64
12  wind_degree                   123356 non-null  int64
13  wind_direction                123356 non-null  object
14  pressure_mb                   123356 non-null  int64
15  pressure_in                   123356 non-null  float64
16  precip_mm                     123356 non-null  float64
17  precip_in                     123356 non-null  float64
18  humidity                      123356 non-null  int64
19  cloud                         123356 non-null  int64
20  feels_like_celsius            123356 non-null  float64
21  feels_like_fahrenheit         123356 non-null  float64
22  visibility_km                 123356 non-null  float64
23  visibility_miles              123356 non-null  int64
24  uv_index                      123356 non-null  float64
25  gust_mph                      123356 non-null  float64
26  gust_kph                      123356 non-null  float64
27  air_quality_Carbon_Monoxide   123356 non-null  float64
28  air_quality_Ozone             123356 non-null  float64
29  air_quality_Nitrogen_dioxide  123356 non-null  float64
30  air_quality_Sulphur_dioxide   123356 non-null  float64
31  air_quality_PM2.5             123356 non-null  float64
32  air_quality_PM10              123356 non-null  float64
33  air_quality_us-epa-index      123356 non-null  int64
34  air_quality_gb-defra-index    123356 non-null  int64
35  sunrise                       123356 non-null  object
36  sunset                        123356 non-null  object
37  moonrise                      123356 non-null  object
38  moonset                       123356 non-null  object
39  moon_phase                    123356 non-null  object
40  moon_illumination             123356 non-null  int64
41  year                          123356 non-null  int32
42  month                         123356 non-null  int32
43  day                           123356 non-null  int32
```

CHAPTER 3: MILESTONE 2 – CORE ANALYTICAL INSIGHTS & INTERACTIVE VISUALIZATION

3.1 Objective of Milestone 2

The second phase of the ClimateScope project shifted the Focus from data cleaning to deep-tier analytical exploration. The primary objective was to move beyond static records and uncover the "Scientific Narrative" hidden within the global weather variables. This milestone was designed to:

- Establish Statistical Baselines: Define what constitutes "normal" versus "anomaly" using descriptive modelling.
- Map Atmospheric Dependencies: Quantify the relationship between thermal, moisture, and pressure variables using correlation matrices.
- Isolate Extreme Events: Implement a mathematical filter using percentiles to identify heat and cold waves.
- Prototype Interactive Intelligence: Transition the project into a dynamic user interface using Streamlit and Plotly.

3.2 Statistical Analysis: Distributions and Temporal Trends

The analytical engine for this milestone was built using Pandas, Matplotlib, and Seaborn. We initiated a multi-layered statistical profile to understand the variance in global weather patterns.

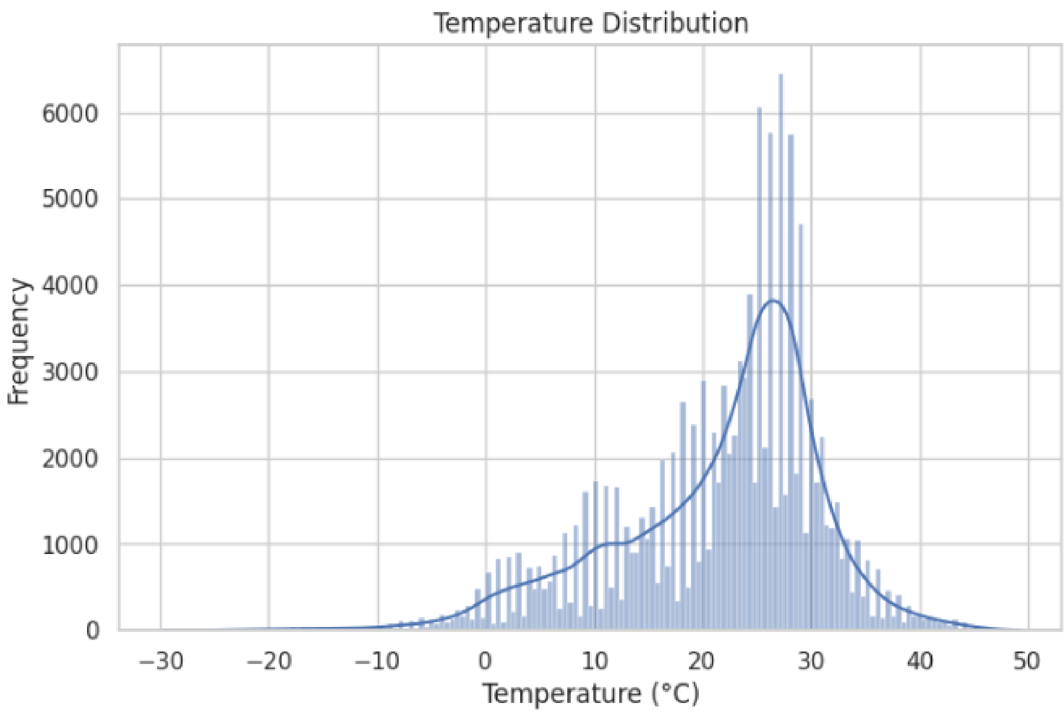
3.2.1 Descriptive Profiling

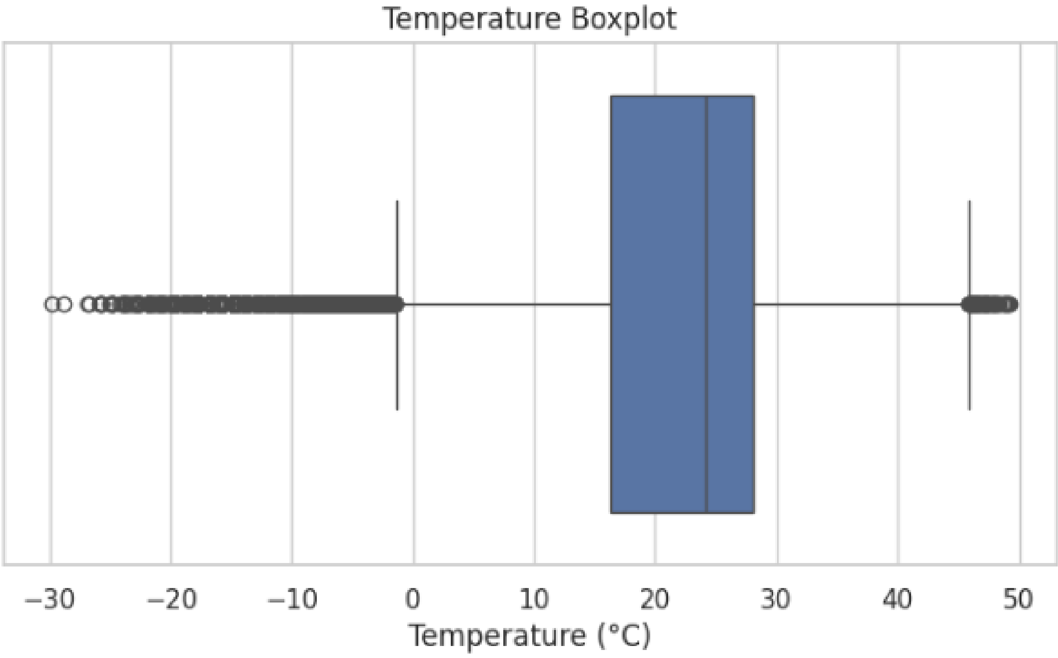
By applying the .describe() function to specific subsets including temperature_celsius, humidity, wind_kph, and pressure_mb, we established the mean and standard deviation for the global climate. This provided the mathematical boundaries required to identify true weather extremes.

	temperature_celsius	humidity	wind_kph	pressure_mb
count	123356.000000	123356.000000	123356.000000	123356.000000
mean	21.665486	66.061383	12.995777	1014.085428
std	9.547590	24.028847	11.891706	10.627979
min	-29.800000	2.000000	3.600000	947.000000
25%	16.300000	50.000000	6.500000	1010.000000
50%	24.100000	71.000000	11.200000	1014.000000
75%	28.100000	85.000000	17.600000	1018.000000
max	49.200000	100.000000	2963.200000	3006.000000

3.2.2 Temperature Distribution and Outlier Detection

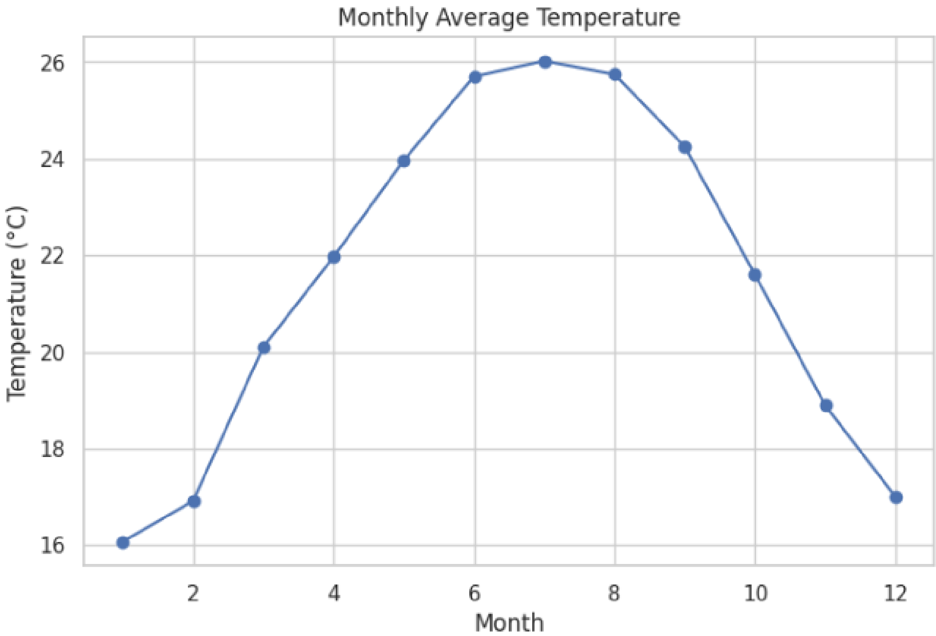
To visualise the spread of global temperatures, we utilised a Histogram with a Kernel Density Estimate (KDE) and a Boxplot. These visuals allowed us to identify the central tendency of the weather data and detect any statistical outliers that fall outside the interquartile range.

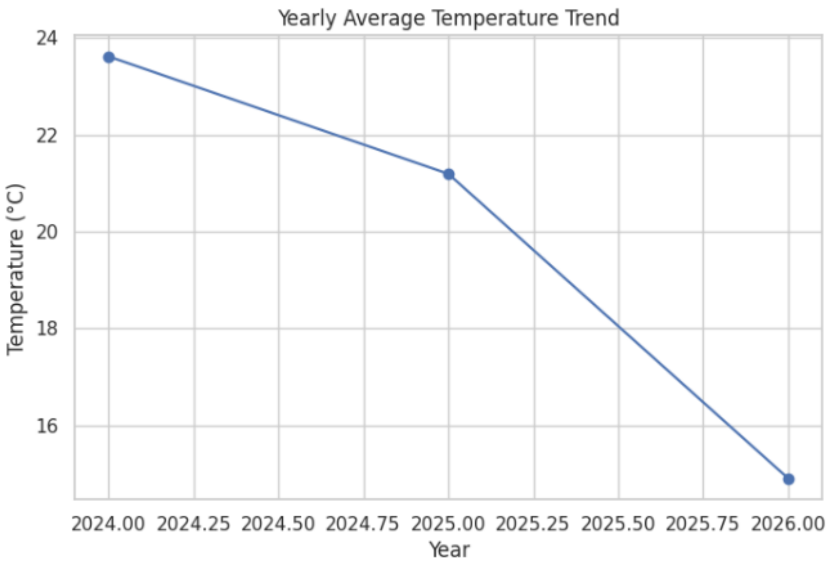




3.2.3 Temporal and Seasonal Trend Mapping

To identify seasonal fluctuations, the data were aggregated to calculate monthly and yearly averages using the .groupby() and .agg() functions. This step was crucial for observing how global temperatures transition between seasons and identifying yearly shifts in thermal intensity.



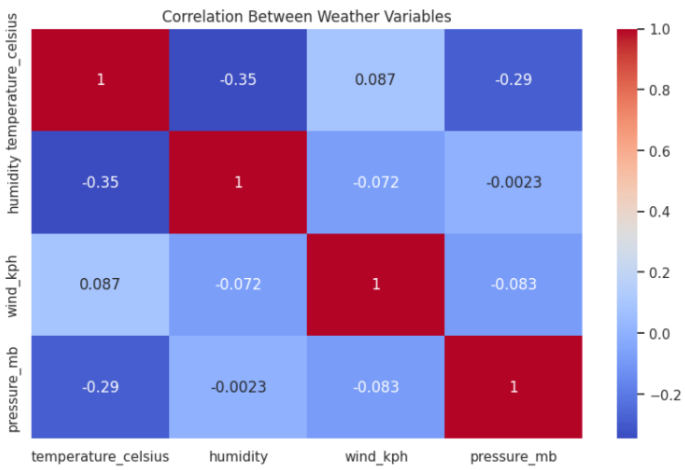


3.3 Correlation Analysis: Mapping Inter-Variable Relationships

A central component of Milestone 2 was examining how weather variables influence one another. We executed a correlation matrix to identify positive and negative dependencies.

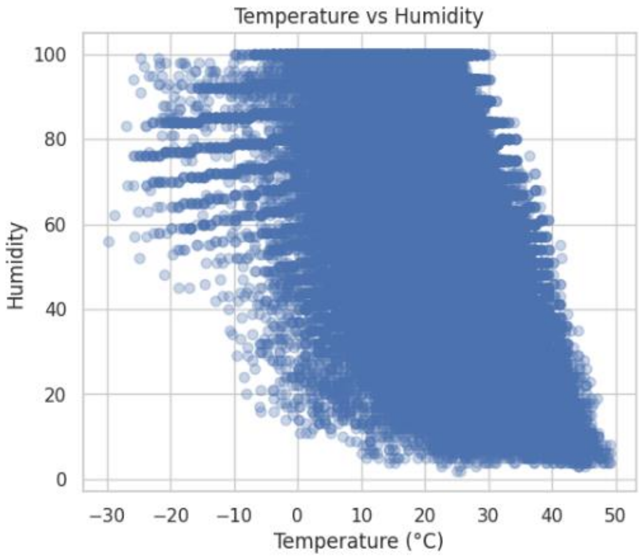
3.3.1 Correlation Heatmapping

To make the numerical relationship tables accessible, we utilised a Seaborn Heatmap. This allowed us to quickly identify "Climate Clusters" where certain variables, such as humidity and temperature, showed a strong negative or positive correlations.



3.3.2 Relationship Analysis

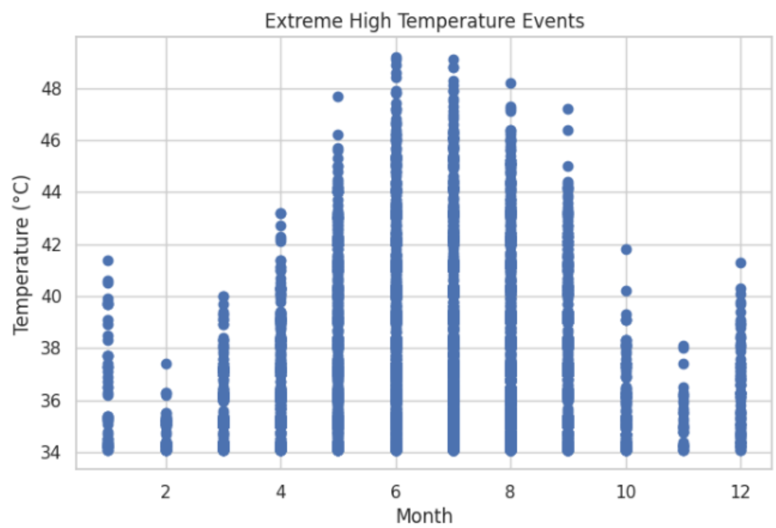
To dive deeper into specific dependencies, we generated Scatterplots (e.g., Temperature vs. Humidity). This provided a visual representation of how moisture levels fluctuate in relation to thermal changes across the global dataset.



3.4 Extreme Weather Events: The Percentile-Based Approach

To move beyond subjective definitions of weather, this project implemented a rigorous percentile-based statistical approach. By calculating the 95th and 5th percentiles, we created a mathematical filter to isolate hazardous events:

- Heat Wave Identification (95th Percentile): Any temperature record exceeding this threshold was flagged as a high-risk thermal event.
- Cold Wave Identification (5th Percentile): Any record falling below this threshold was categorised as an extreme cold event.

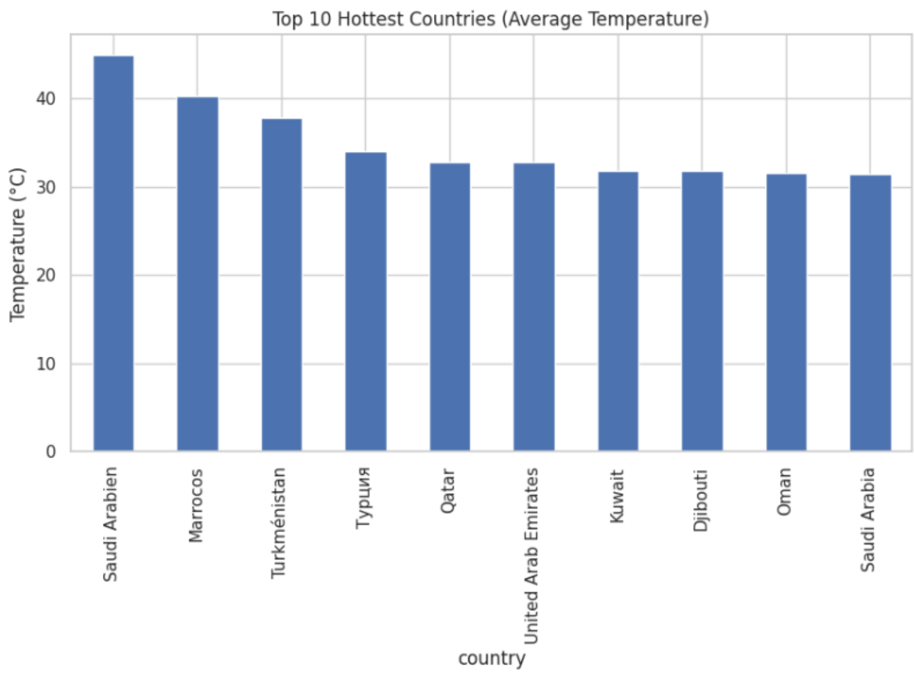


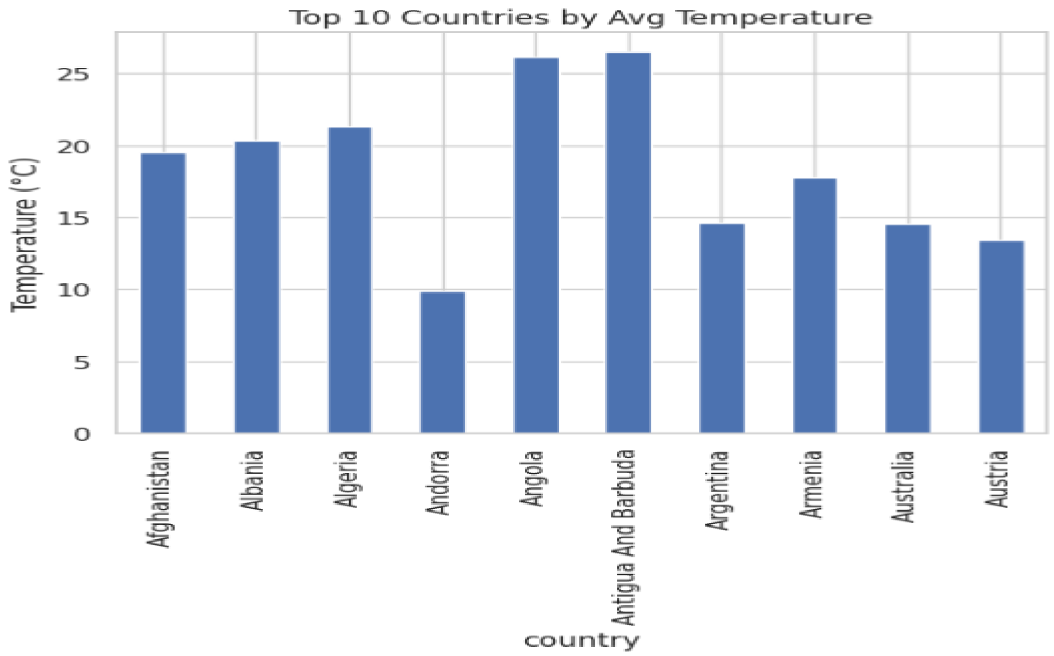
3.5 Regional Comparison and Geospatial Distribution

The analysis then scaled from global averages to country-specific comparisons. By grouping the dataset by the country attribute, we identified the hottest and coolest regions in the repository.

3.5.1 Top 10 Hottest Countries

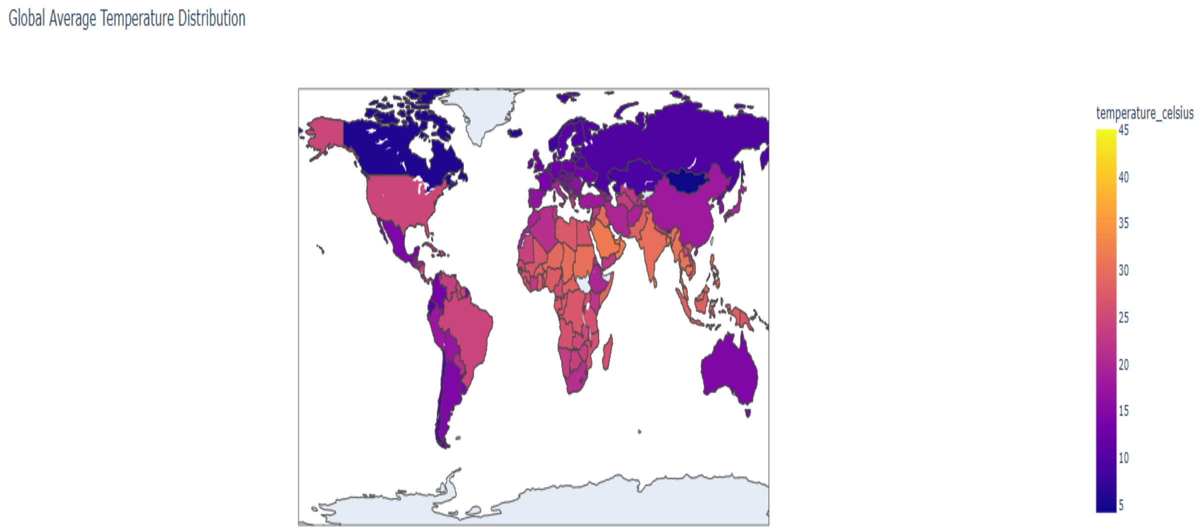
A bar chart was utilised to rank countries by their average temperature, identifying regions that consistently face high thermal stress.





3.5.2 Global Average Temperature Distribution

Finally, we utilised **Plotly Express** to generate a **Choropleth Map**. This geographic visualisation provides a global bird's-eye view, mapping temperature intensity directly onto a world map for immediate spatial analysis.



CHAPTER 4: MILESTONE 3 – VISUALIZATION & DASHBOARD ARCHITECTURE

4.1 Objective of Milestone 3

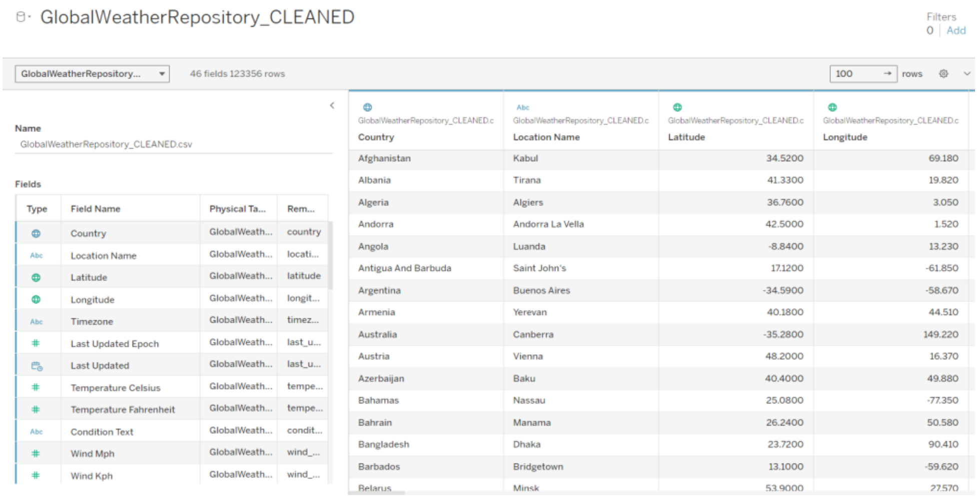
The final milestone of the ClimateScope project was dedicated to "Visual Intelligence." The goal was to Synthesise the 43 variables and the analytical findings from Milestone 2 into a single, synchronised dashboard. The objectives included:

- Interactive Synthesis: Linking multiple worksheets so they react to a single user filter.
- Geospatial Intelligence: Creating a high-detail map to visualise weather intensity by location.
- KPI Implementation: Displaying real-time averages for temperature, wind, and air quality.
- User-Centric Design: Developing a "Bento Box" layout that organises complex data into digestible modules.

4.2 Data Integration and Connection

The process began by importing the final GlobalWeatherRepository_CLEANED.csv into Tableau Desktop.

- Data Interpreter: Tableau's built-in interpreter was used to ensure that the 2023–2025 date fields and geographic names (Country/City) were recognised with 100% accuracy.
- Variable Categorisation: Variables were separated into Dimensions (Country, Condition, Year) and Measures (Temperature, Humidity, PM2.5) to facilitate drag-and-drop visualisation.

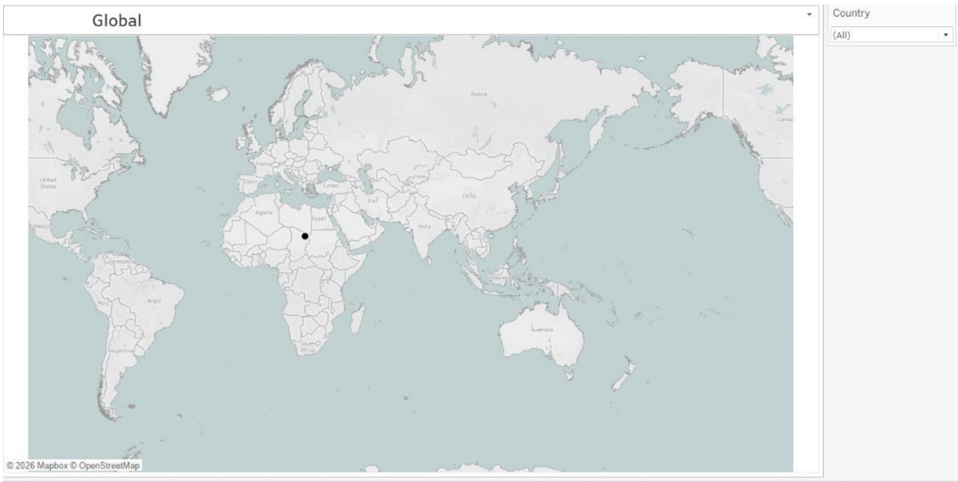


4.3 Worksheet Development (The Building Blocks)

Before the final dashboard was assembled, individual Worksheets were created to represent different climates dimensions:

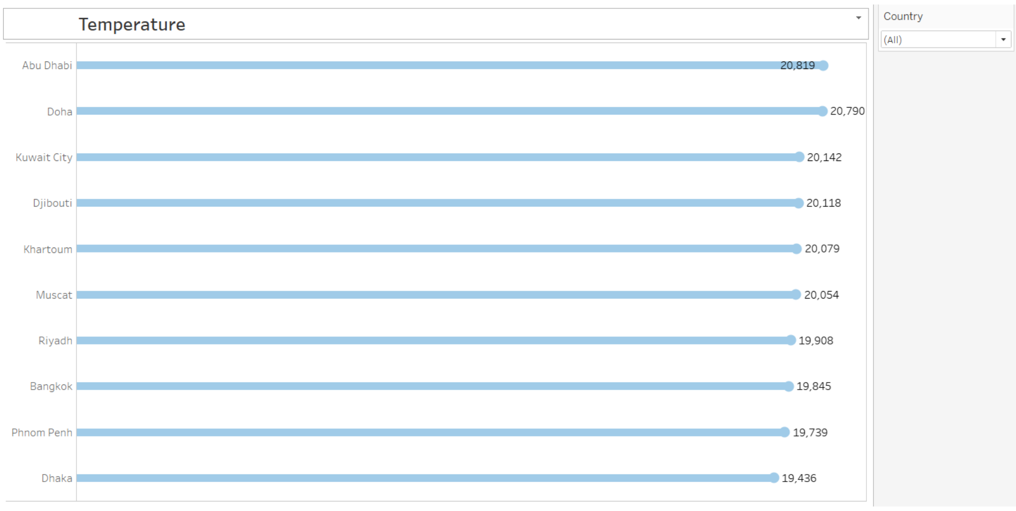
4.3.1 Global Weather Map (Geospatial View)

Using the latitude and longitude data, a symbol map was generated. The size of the marks was tied to temperature_celsius, while the colour represented the condition_text. This allows users to spot immediately "Heat Corridors."



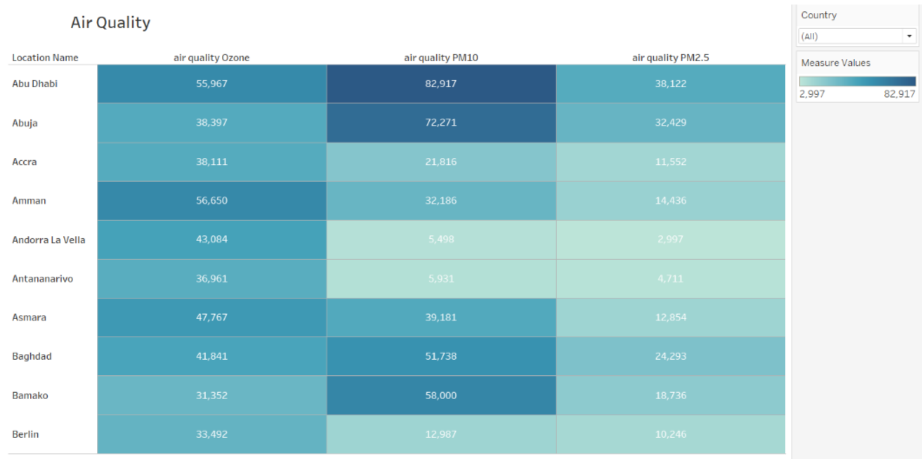
4.3.2 Environmental Variable Trends (Line Charts)

Leveraging the year and month features engineered in Milestone 1: We created time-series line charts to track the progression of humidity and temperature over the 2023–2025 timeline.



4.3.3 Air Quality Matrix (Tree Map/Bar Chart)

A dedicated module for Particulate Matter and UV index was created to highlight public health risks. This visualisation ranks cities based on their pollution levels.

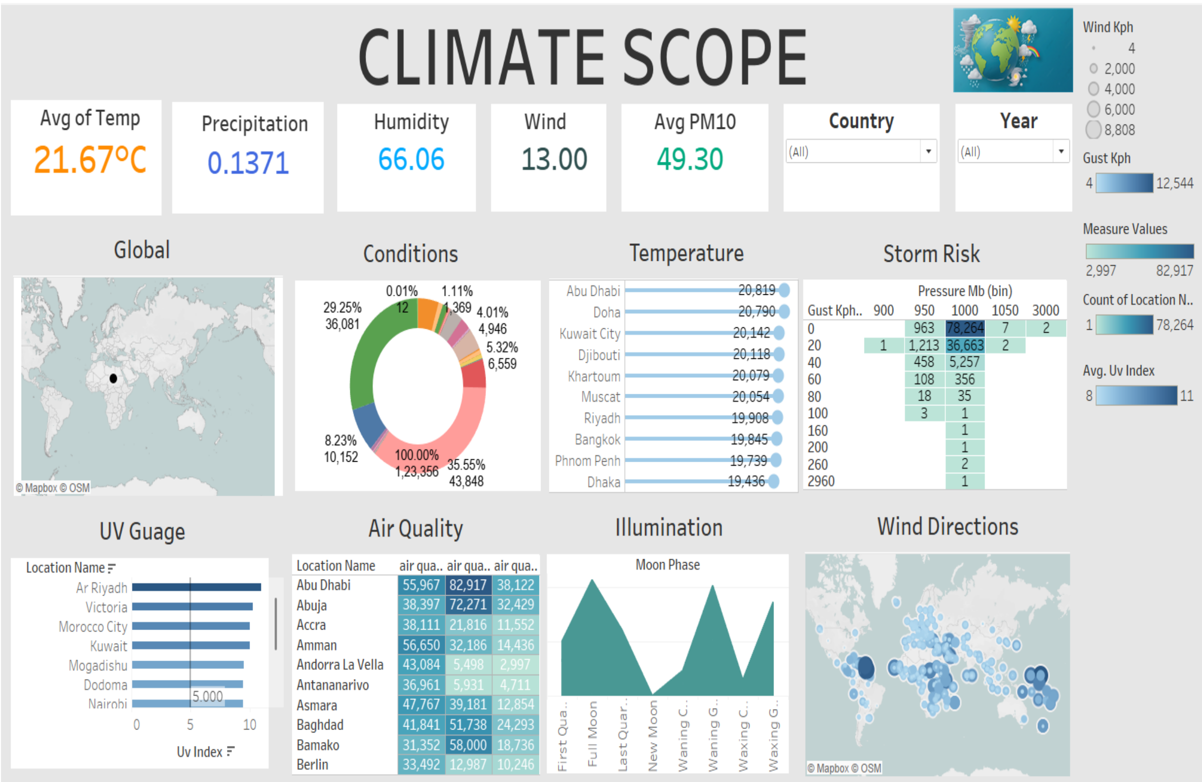


4.4 Dashboard Architecture: The "Bento Box" Design

The final dashboard was constructed using a tiled "Bento Box" layout, ensuring that the most critical Information is visible without scrolling.

- **Synchronisation (Actions)**: "Filter Actions" were applied so that clicking on a specific country on the map automatically updates the line charts and air quality metrics for that region.
- **KPI Header**: At the top of the dashboard, four large-text "Big Number" (BAN) tiles were placed to show:

1. Avg Global Temp
2. Max Wind Gust
3. Avg Humidity
4. Air Quality Index Peak



4.5 Dashboard Interactivity and Use Cases

The completed ClimateScope platform supports several professional use cases:

1. Anomaly Detection: A researcher can filter for "Extreme" conditions to see where heat waves are occurring.
2. Regional Comparison: Users can compare the air quality of industrial cities versus rural locations.
3. Temporal Analysis: Decision-makers can observe if humidity levels have increased year-over-year.

CONCLUSION

The ClimateScope project successfully transitions from raw data collection to a sophisticated visual intelligence platform. By leveraging the Global Weather Repository (2023–2025), the project moved through a rigorous lifecycle of data cleaning in Google Colab, statistical exploration in Python, and high-fidelity dashboarding in Tableau.

The technical execution proved that high-density environmental variables—such as Particulate Matter, UV Index, and Barometric Pressure—can be synthesised into an interactive "Bento Box" architecture that is accessible to both researchers and public health officials. Through the implementation of percentile-based anomaly detection and geographic correlation mapping, the project meets its primary objective: providing a clear, data-driven window into the complexities of global climate volatility.

This internship project stands as a complete proof-of-concept for how modern data engineering and BI tools can be synchronised to turn massive datasets into actionable climate insights.